THE FUNCTOR F_! COMMUTES WITH DIRECT SUMS FOR SEPARATED MORPHISMS LOCALLY OF FINITE TYPE

THE STACKS PROJECT AUTHORS

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Editorial note. Editorial note: the final step is left to the reader (looking at supports of sections); it is a definitional identification using the definition of f_! as the subsheaf of sections with proper support together with the displayed equality H^0(X, F) = direct sum H^0(X, F_i) from the colimit theorem.

Allowed standard prerequisites (editorial).

- Colimit description of f_! for separated morphisms (Lemma lemma-colim-f-shriek-separated, canonical tag 0F54). cited at proof lines 969-971. Reduces the claim from arbitrary X to quasi-compact X.
- Colimit theorem for sheaves (Etale Cohomology, Theorem etale-cohomology-theorem-colimit, canonical tag 09YQ). cited at proof lines 972-973. Gives H^0(X, F) = direct sum H^0(X, F_i) for quasi-compact X.

Source reference. Original lemma 0F54: sources/stacks-chow/tags.txt, source lines 18924–18924. Cited at proof lines 969-971: reduces the claim from arbitrary X to quasi-compact X.

Source reference. Original theorem 09YQ: sources/stacks-chow/tags.txt, source lines 12573–12573. Cited at proof lines 972-973: gives H^0(X, F) = direct sum H^0(X, F_i) for quasi-compact X.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes which is separated and locally of finite type. Then functor $f_!$ commutes with direct sums.*

Proof. Let $\mathcal{F} = \bigoplus \mathcal{F}_i$. To show that the map $\bigoplus f_!\mathcal{F}_i \to f_!\mathcal{F}$ is an isomorphism, it suffices to show that these sheaves have the same sections over a quasi-compact object V of $Y_{étale}$. Replacing Y by V it suffices to show $H^0(Y, f_!\mathcal{F}) \subset H^0(X, \mathcal{F})$ is equal to $\bigoplus H^0(Y, f_!\mathcal{F}_i) \subset \bigoplus H^0(X, \mathcal{F}_i) \subset H^0(X, \bigoplus \mathcal{F}_i)$. In this case, by writing X as the union of its quasi-compact opens and using Lemma 0F54 we reduce to the case where X is quasi-compact as well. Then $H^0(X, \mathcal{F}) = \bigoplus H^0(X, \mathcal{F}_i)$ by Étale Cohomology, Theorem 09YQ. Looking at supports of sections the reader easily concludes. □

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